

Hakan Tugrul Otal

PhD Student & Cybersecurity-AI Researcher
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PhD Student and Research Assistant specializing in cybersecurity and artificial intelligence. My research bridges the threats and risks of Generative AI with robust system defense, combining hands-on experience in secure software development with academic and industrial projects focused on protecting critical infrastructures and data assets.

INTERESTS

Cybersecurity, Cyber Risk, AI/ML, Generative AI, Data Privacy, Natural Language Processing

EDUCATION

University at Albany, SUNY , New York, United States Ph.D. in Information Science Data Analytics (Major) – Cybersecurity (Minor)	08-2023 — Present GPA: 3.9/4.0
University at Albany, SUNY , New York, United States M.S. in Information Science Artificial Intelligence and Data Analytics	08-2023 — 05-2026 GPA: 3.9/4.0
Istanbul Technical University , Istanbul, Turkey B.S. in Computer Science	09-2018 — 06-2023 GPA: 3.5/4.0

EXPERIENCE

University at Albany, SUNY <i>Graduate Research Assistant</i>	08-2023 — Present
• Conducted research on Gen-AI security and safety at TATAR Lab under the supervision of Dr. Unal Tatar. • Conducted research on integrating AI/LLM techniques for cyber threat detection at AI in Complex Systems Lab under the supervision of Dr. M. Abdullah Canbaz. • Supported various initiatives as a Graduate Assistant for Public Engagement and Enrollment. • Presented research at various conferences and workshops. • Teaching Assistant for CINF135: Concepts of AI.	
CyRisk Inc. <i>AI/Software Engineer Intern</i>	12-2025 — 05-2026
• Developed AI features for a cyber risk engine, including automated risk score calculation and analysis. • Worked on cybersecurity litigation cases leveraging AI-driven insights for risk assessment. • Utilized AWS, Docker, and Postman for development, deployment, and API testing workflows.	
TATAR Lab (University at Albany, SUNY) <i>Summer Research Intern</i>	05-2025 — 08-2025
• Worked on research projects related to AI security and safety. • Supervised student interns and led research group of students from VICEROY.	
AI in Complex Systems Lab (University at Albany, SUNY) <i>Summer Research Intern</i>	05-2024 — 08-2024
• Worked on two research projects and published conference papers from them. • Supervised undergraduate interns at UAlbany Summer Research Program (UASRP).	
Parieta NLP <i>Software Engineer</i>	01-2023 — 07-2023

- Designed and implemented the full system architecture for frontend, backend, AI, and database.
- Developed the web application UI and integrated e-commerce features.
- Developed Nest.js backend services, MongoDB databases, and AI pipelines for the application.
- Deployed and maintained Docker containers on AWS infrastructure.

Deka Technology

Software Engineer

07-2022 — 12-2022

- Developed new API endpoints for enterprise-scale applications.
- Connected SAP services to backend systems through a customized integration layer.
- Worked with 100+ microservices using Jenkins, Openshift, Elastic, Redis, Quarkus, Apache Camel, and Keycloak.

PoiLabs

Software Engineer Intern

09-2021 — 10-2021

- Developed backend services for a mobile application for visually impaired people.
- Refactored the mobile application API with more dynamic and efficient approaches.
- Collaborated effectively using Bitbucket as VCS in a team setting.

SKILLS

- **Programming:** Python, C/C++, Javascript/Typescript
- **ML Frameworks:** Pytorch, Tensorflow, Sklearn
- **Security and Network Tools:** Wireshark, Nmap, Neo4j, Gephi, Networkx, PyG
- **Data Visualization & Manipulation:** Matplotlib, Seaborn, Numpy, Pandas
- **Database Systems:** MongoDB, PostgreSQL, MySQL
- **MLOps & Cloud:** Docker, Elasticsearch, Redis, AWS, GCloud
- **LLM Techs:** Huggingface, Ollama, LMStudio, Unsloth, Llama-Factory

PUBLICATIONS

- [1] M Abdullah Canbaz, Hakan Otal, Tugce Unlu, Nour Alhussein, and Brian Nussbaum. **Who Coordinates US Cyber Defense? A Co-Authorship Network Analysis of Joint Cybersecurity Advisories (2024–2025)**. *arXiv preprint arXiv:2510.20080*, 2025.
- [2] M Mikail Demir, Hakan T Otal, and M Abdullah Canbaz. **LegalGuardian: A Privacy-Preserving Framework for Secure Integration of Large Language Models in Legal Practice**. *arXiv preprint arXiv:2501.10915*, 2025.
- [3] Hao Jian Huang, Hakan T Otal, and M Abdullah Canbaz. **Federated learning in adversarial environments: Testbed design and poisoning resilience in cybersecurity**. In **2025 IEEE International Conference on Communications Workshops (ICC Workshops)**, pages 1079–1084. IEEE, 2025.
- [4] Hakan T Otal and M Abdullah Canbaz. **Ai-powered crisis response: Streamlining emergency management with llms**. In **2024 IEEE World Forum on Public Safety Technology (WFPST)**, pages 104–107. IEEE, 2024.
- [5] Hakan T Otal and M Abdullah Canbaz. **Llm honeypot: Leveraging large language models as advanced interactive honeypot systems**. In **2024 IEEE Conference on Communications and Network Security (CNS)**, pages 1–6. IEEE, 2024.
- [6] Hakan T Otal, Stephen V Faraone, and M Abdullah Canbaz. **A New Perspective on ADHD Research: Knowledge Graph Construction with LLMs and Network Based Insights**. In **International Conference on Complex Networks and Their Applications**, pages 337–349. Springer Nature Switzerland Cham, 2024.
- [7] Hakan T Otal, Eric Stern, and M Abdullah Canbaz. **Llm-assisted crisis management: Building advanced llm platforms for effective emergency response and public collaboration**. In **2024 IEEE Conference on Artificial Intelligence (CAI)**, pages 851–859. IEEE, 2024.

- [8] Hakan T Otal, Abdulhamit Subasi, Furkan Kurt, M Abdullah Canbaz, and Yasin Uzun. **Analysis of gene regulatory networks from gene expression using graph neural networks**. In **Digital Healthcare, Digital Transformation and Citizen Empowerment in Asia-Pacific and Europe for a Healthier Society**, pages 249–270. Academic Press, 2025.
- [9] Hakan T Otal, Elyse Zavar, Sherri B Binder, Alex Greer, and M Abdullah Canbaz. **Harnessing Deep Learning and Satellite Imagery for Post-Buyout Land Cover Mapping**. In **2024 IEEE Conference on Artificial Intelligence (CAI)**, pages 665–671. IEEE, 2024.

PROJECTS

Autonomous Cyber-Offensive LLM Agents (VICEROY Project) 2026

- Leveraged Vertex AI to deploy large-scale, autonomous agents using a hybrid of frontier models and fine-tuned cybersecurity open-source models on Google Cloud infrastructure.
- Advanced the D-CIPHER multi-agent framework and contributed to the autonomous Cyber-Offensive LLM Agent research created by the NYUAD Center for Cyber Security.
- Authored a research paper for presentation to DoW cyber spectrum leaders at the VICEROY Symposium.
- Supervised by Dr. Unal Tatar.

ATTASH - Anticipating Terrorist Technology Adoption to Secure the Homeland 2024

- Contributed to the ATTASH Tool Pilot as part of the development team.
- Developed Monte Carlo simulations in Excel and supporting Python scripts for statistical analysis.
- Supervised by Dr. Gary Ackerman.

Distributed Network Load Generation Software (B.S. Graduation Project) 2023

Trust Based Data Storage for E-Health Systems in IoT (B.S. Lab Project) 2021

SERVICES

IEEE

IEEE Student Member	09-2023 — Present
IEEE Computational Intelligence Society (CIS) Member	03-2024 — Present

ACM

Student Member	10-2024 — Present
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University at Albany, SUNY

Member of AI in Complex Systems Lab	08-2023 — Present
Member of TATAR Lab	05-2025 — Present
Graduate Assistant of Assistant Dean for Public Engagement and Enrollment	Spring 2025
Supervisor of Student Interns and VICEROY Research Group at TATAR Lab	Summer 2025
Supervisor of Undergrad. Interns at UAlbany Summer Research Program (UASRP)	Summer 2024
Teaching Assistant for ‘CINF135: Concepts of AI’	Spring 2024, Fall 2024

AWARDS & GRANTS

SUNY AI Platform – Google Cloud

Cloud computing grant (\$30K) for advanced AI model access and compute resources	2025 — 2026
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IEEE Computational Intelligence Society

Student Travel Grant for IEEE CAI (\$1200)	2024
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IEEE Communications Society

Student Travel Grant for IEEE CNS (\$800)	2023, 2024
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University at Albany, SUNY

Ph.D. Full Tuition Scholarship and Stipend	2023 — Present
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PRESENTED

NTIR Workshop - "Building a Chatbot with RAG from Scratch"	2026
ACM UAlbany Workshop - "Deploying Models on the Edge"	2025
NTIR (New Trends in Informatics Research Conference)	2024, 2025
IEEE International Conference on Communications Workshops (ICC)	2025
International Conference on Complex Networks and Their Applications (CNA)	2024
IEEE Conference on Artificial Intelligence (CAI)	2024
IEEE World Forum on Public Safety Technology (WFPST)	2024
IEEE Conference on Communications and Network Security (CNS)	2023, 2024

REVIEWED

Conference on Computational Linguistics (COLING)	2025
Information Security Journal: A Global Perspective	2024
IEEE Conference on Communications and Network Security (CNS)	2024
IEEE Conference on Local Computer Networks (LCN)	2024
IEEE International Conference on Computer Communications (INFOCOM)	2024
Complex Networks and their Applications (CNA)	2023, 2024

Recommendation Letters Available upon Request:

Dr. Unal Tatar

Associate Professor

Department of Cybersecurity

University at Albany SUNY, NY, USA

E-mail: utatar@albany.edu

Website: albany.edu/cehc/faculty/unal-tatar

Dr. M Abdullah Canbaz

Assistant Professor

Department of Information Sciences and Technology

University at Albany SUNY, NY, USA

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Dr. Tolga Ovatman

Associate Professor

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